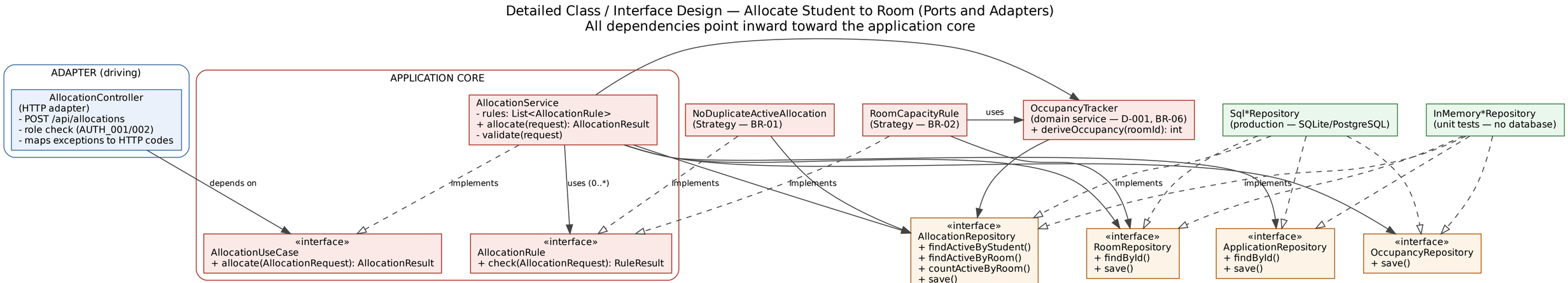
Detailed Class / Interface Design — Allocate Student to Room (Ports and Adapters)
All dependencies point inward toward the application core
ADAPTER (driving)
AllocationController
(HTTP adapter)
- POST /api/allocations
- role check (AUTH_001/002)
- maps exceptions to HTTP codes
APPLICATION CORE
AllocationService
- rules: List<AllocationRule>
+ allocate(request): AllocationResult
- validate(request)
NoDuplicateActiveAllocation
(Strategy — BR-01)
RoomCapacityRule
(Strategy — BR-02)
uses
OccupancyTracker
(domain service — D-001, BR-06)
+ deriveOccupancy(roomId): int
Sql*Repository
(production — SQLite/PostgreSQL)
InMemory*Repository
(unit tests — no database)
depends on
implements
uses (0..*)
implements
implements
implements
implements
«interface»
AllocationUseCase
+ allocate(AllocationRequest): AllocationResult
«interface»
AllocationRule
+ check(AllocationRequest): RuleResult
«interface»
AllocationRepository
+ findActiveByStudent()
+ findActiveByRoom()
+ countActiveByRoom()
+ save()
«interface»
RoomRepository
+ findById()
+ save()
«interface»
ApplicationRepository
+ findById()
+ save()
«interface»
OccupancyRepository
+ save()